I ran Qwen 7B parameter model on BIRD Mini Dev with 500 questions. First I did it without schema context. This way, I had a result like this:

```
                    simple              moderate            challenging         total
count               148                 250                 102                 500
====================================    EX    ====================================
EX                  3.38                1.20                0.98                1.80
========================================================================================
```

Then I added table definitions, column names and a schema summary. Here is the result for that:

```
                    simple              moderate            challenging         total
count               148                 250                 102                 500
====================================    EX    ====================================
EX                  49.32               31.60               18.63               34.20
========================================================================================
Finished EX evaluation for SQLite on Mini Dev set
```

Terminal prompts to make it work:

1- No Schema Context

python bird_infer.py

python evaluation/evaluation_ex.py --predicted_sql_path predict_mini_dev.json --ground_truth_path data/mini_dev_sqlite_gold.sql --db_root_path mini_dev_data/dev_databases/minidev/MINIDEV/dev_databases/ --diff_json_path data/mini_dev_sqlite.json

2- With Schema Context

python bird_infer_schema.py

python evaluation/evaluation_ex.py --predicted_sql_path predict_mini_dev_schema.json --ground_truth_path data/mini_dev_sqlite_gold.sql --db_root_path mini_dev_data/dev_databases/minidev/MINIDEV/dev_databases/ --diff_json_path data/mini_dev_sqlite.json

3- With fuzzy string and BM25 (vol1)

```
h data/mini_dev_sqlite_gold.sql --db_root_path mini_dev_data/dev_databases/minidev/MINIDEV/dev_databases/ --diff_json_path data/mini_dev_sqlite.json
start calculate EX
                    simple              moderate            challenging         total
count               148                 250                 102                 500
====================================    EX    ====================================
EX                  22.97               23.20               14.71               21.40
========================================================================================
Finished EX evaluation for SQLite on Mini Dev set
```

Problems:

Pruned Bridge Tables (Broken Foreign Key Paths): When a question asks about customers and loans, BM25 and fuzzy matching assign high scores to those two tables. However, in

relational schemas, connecting customers to loans often requires a bridge table (e.g., account or disp). Because the bridge table's name is not explicitly mentioned in the question or evidence, the pruner discarded it. Without it, SQLite cannot resolve the join.

Column Limit (top_k = 8): Limiting columns per table caused columns used in WHERE, ORDER BY, or GROUP BY clauses to be eliminated if their names didn't match the query keywords directly. When the model tried to reference them, SQLite threw no such column.

Database Size Mismatch: BIRD Mini-Dev databases have only 3 to 8 tables on average. Qwen 2.5 7B easily fits their entire schemas into context (~2K tokens). Pruning tables on small databases created an artificial information bottleneck with no context-window benefit.

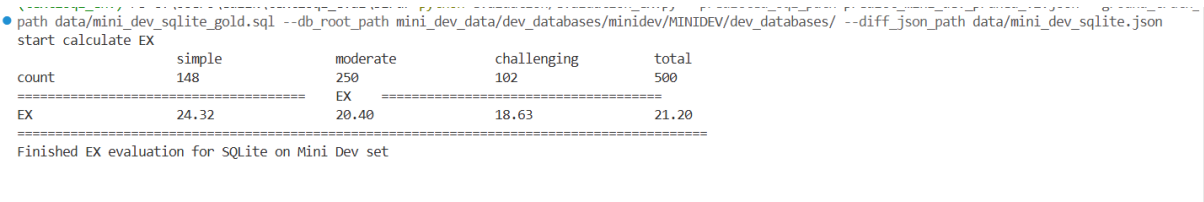
4- Fuzzy string and BM25 (vol2)

Changes:

T**able Retention / FK Graph Expansion:** If a database has ≤ 6 tables, keep all of them. For larger databases, find seed tables using BM25/Fuzzy, then compute the shortest paths over the foreign-key graph to automatically restore any intermediate bridge tables.

Join-Key Preservation: All primary keys and foreign keys belonging to selected tables are strictly preserved.

Relaxed Column Selection: Instead of a hard cap of 8, keep all relevant columns plus any column scoring above a minimum threshold, ensuring essential attributes aren't discarded.



Simple and challenging questions increased but moderate went down, total is almost the same. Possible causes:

1- In both pruned scripts, the prompt explicitly demands: Output ONLY the SQL statement on a single line without explanations or markdown. BIRD questions are multi-step reasoning problems that require calculating ratios, handling conditional edge cases, casting data types, and filtering intermediate groupings. When forced to output the SQL immediately, small parameter models like Qwen 2.5 7B cannot allocate intermediate reasoning tokens. (WE HAVEN'T TRIED OUR MULTIPLE PASS PROPOSAL YET)

2- BIRD databases frequently use cryptic categorical codes. Without 2–3 example values per column the model hallucinates exact string literals like WHERE status = 'Approved', resulting in empty result sets and an automatic 0 EX for that question. Examples and maybe database dictionary?

3- Table amount. There are 3-11 tables per database so using these might not be that affective. Here (maybe) pruning provides minimal token reduction while risking the accidental omission of secondary filter columns.